Northwestern Medicine

Evaluation of Local Language Models for Medical Record Analysis

A comparative evaluation of Qwen, Llama, and Gemini models using PDSQI-9 based quality assessment.

Presented by: Cherish Uwandu

Collaborating Authors: David Liebovitz, MD

Department: Clinical Informatics

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Introduction

- **Medical records are information-dense.** Patients' records contain large amounts of longitudinal and often complex clinical information. [1]
- **Large language models (LLMs) may improve access to this information.** They are being investigated for tasks including medical question-answering, summarization, and clinical reasoning. [1]
- **Local LLMs offer potential advantages.** Processing information directly on a device may improve privacy and reduce reliance on cloud-based services. [2]
- **However, smaller models require careful evaluation.** Healthcare LLMs can produce inaccurate information, making evaluation of response quality and accuracy important before potential clinical use. [1]

Aim

To compare the response quality and accuracy of small local language models against a high-performing reference model when answering questions based on retrieved medical-record data.

Research Question

How do small local language models compare in response quality and accuracy across varying levels of medical-record question difficulty?

Methods: Testing Dataset Creation

Synthetic medical-record data and user questions of varying difficulty were generated using Gemini 3.1 Pro.

Questions and chart contexts were reviewed by a human for relevance, clarity, and answerability.

Each model received the same system prompt, chart data, and user question for all 30 synthetic patient cases, enabling a controlled comparison.

Three local models (Qwen 2.5 1.5B, Qwen 2.5 3B, and Llama 3.2 3B) and a reference model (Gemini 3.1 Pro) generated 120 total responses.

Methods: Model Evaluation

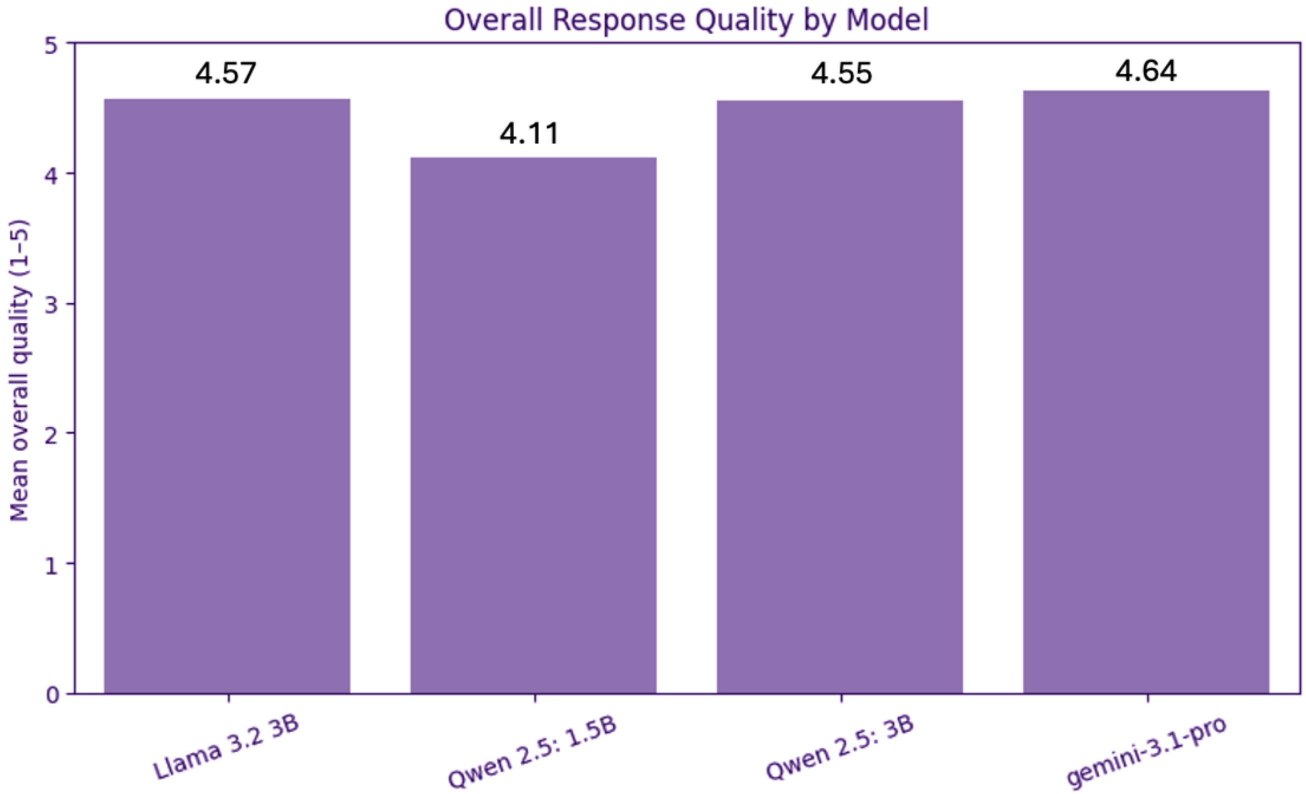
PDSQI-9–Derived Evaluation

- **The Provider Documentation Summarization Quality Instrument (PDSQI-9)**: Framework for evaluating LLM-generated clinical summaries. [3]
  Adapted here for medical-record question answering.
- Claude Sonnet 4.6 evaluated each response using an adapted PDSQI-9 framework and the developers' LLM-as-a-judge implementation.
- **Overall Response Quality**: Mean of 6 dimensions including Accuracy, Thoroughness, Usefulness, Organization, Comprehensibility, and Succinctness.
- The Abstraction and Synthesized dimensions were evaluated separately due to different scoring structures and/or conditional applicability.
- **Excluded dimensions**: Citation (not applicable) and Stigmatizing language (all scores = 0).

Results: Overall Response Quality

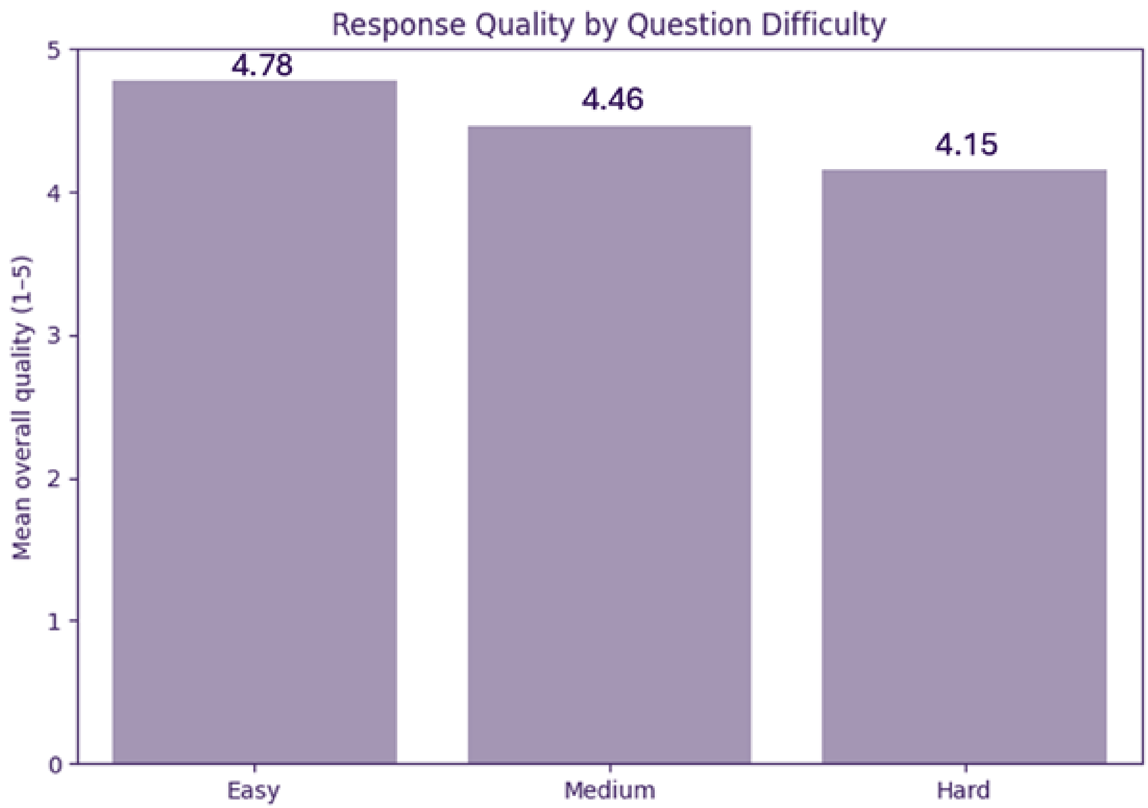
Overall response quality did not significantly differ between models.

(Friedman $\chi^2 = 5.19$, $p = 0.158$).



Results: Effect of Question Difficulty

Response quality significantly differed by question difficulty. *Kruskal-Wallis H = 17.30, p < 0.001*

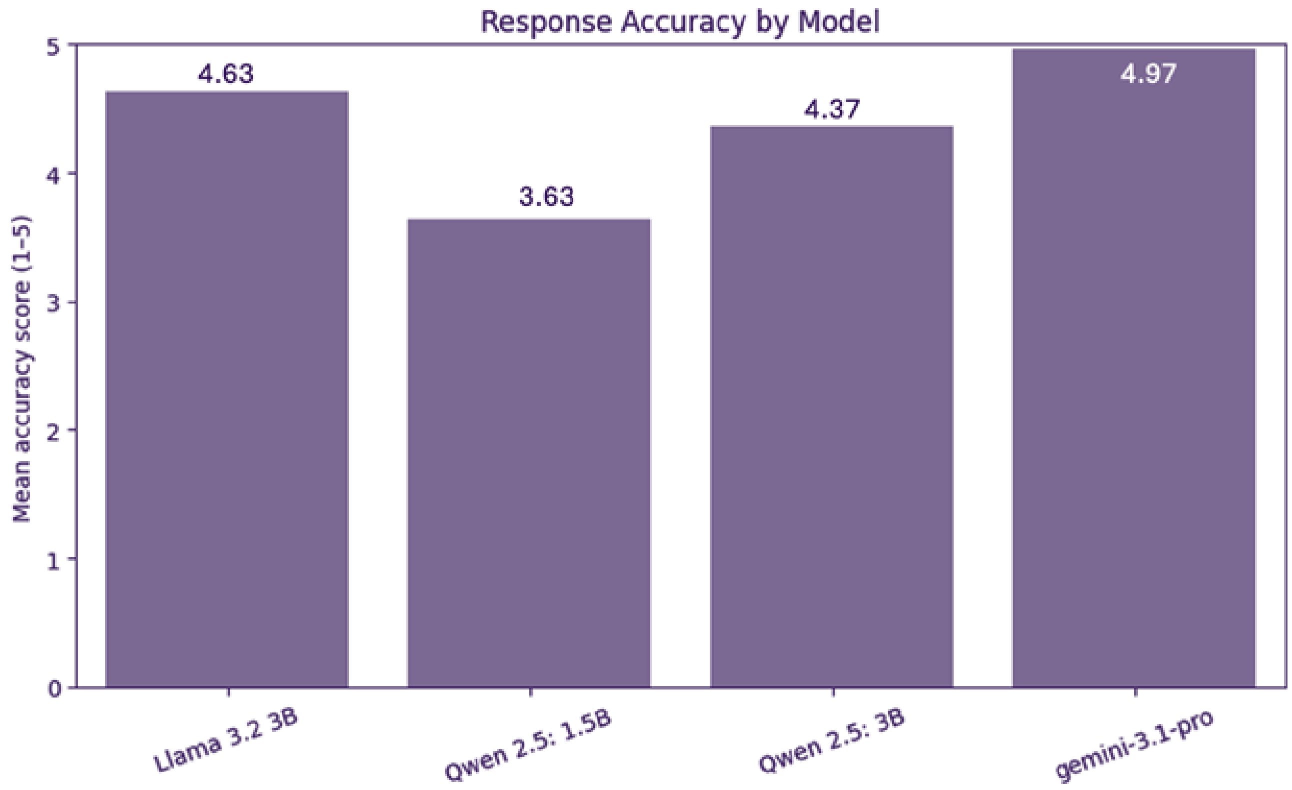


Easy questions scored significantly higher than Medium (Holm-adjusted p = 0.0086) and Hard questions (p = 0.00068). Medium vs. Hard was not significant (p = 0.053).

Results - Accuracy by Model

Accuracy significantly differed between models. *Friedman* $\chi^2 = 23.44, p < 0.001$

The smallest local model had the lowest mean accuracy, while the reference model had the highest.



Qualitative Example

Question: My sleep apnea has been getting worse. Have I gained a significant amount of weight since my CPAP was originally titrated in 2020, and has my daytime oxygen saturation changed?

Model	Response Excerpt	PDSQI-9 Rubric	Accuracy
Qwen 1.5B	“You may experience an increase in snoring and pauses in breathing at night”	Unsupported assertion	2/5
Qwen 3B	“I cannot determine if there was a significant weight gain…”	Misaligned assertion	3/5
Llama 3.2 3B	“You have gained approximately 35 lbs… SpO_2 was 98%… most recent… 94%.”	Chart-supported assertions	5/5
Gemini 3.1 Pro	“Your records track ‘SpO_2’, but they do not explicitly specify whether these readings were taken during the daytime or nighttime.”	Chart-supported; noted missing information	5/5

Relevant chart evidence: OSA diagnosed March 10, 2020; CPAP titration study March 25, 2020. Weight increased from 210 lbs (April 1, 2020) to 245 lbs (December 20, 2023), while SpO_2 decreased from 98% to 94%. The chart does not specify whether SpO_2 readings were daytime or nighttime.

Conclusion

- Local LLMs demonstrated high overall response quality, with no significant difference between models.
- Question difficulty influenced response quality, with Easy questions receiving higher scores than Medium and Hard questions.
- Accuracy differed significantly across models, with Gemini 3.1 Pro performing highest and Qwen 2.5 1.5B lowest.
- Llama 3.2 3B was identified as the candidate local model for subsequent fine-tuning and evaluation.

Future Directions

- Fine-tune the selected local model using the training dataset.
- Re-evaluate the fine-tuned model using the held-out testing dataset.
- Develop condition-specific fine-tuned models that can be swapped on-device.
- Evaluate performance in the context of iPhone-based, local clinical question answering.

References

1. Wang, D., & Zhang, S. (2024). Large language models in medical and healthcare fields: Applications, advances, and challenges. *Artificial Intelligence Review, 57*, 299. https://doi.org/10.1007/s10462-024-10921-0
2. Wang, X., et al. (2025). Empowering edge intelligence: A comprehensive survey on on-device AI models. *ACM Computing Surveys, 57*(9), 228. https://doi.org/10.1145/3724420
3. Croxford, E., et al. (2025). Development and validation of the provider documentation summarization quality instrument for large language models. *Journal of the American Medical Informatics Association, 32*(6), 1050–1060. https://doi.org/10.1093/jamia/ocaf068

GitHub

Scan to access the GitHub repository.

Repository includes:

- Synthetic chart data & evaluation questions
- Model responses and evaluation results
- Statistical analysis and visualization code
- Project documentation and references